

CLÉMENT ROMAC

clement.romac@gmail.com ◇ Bordeaux, France

Age : 29 ◇ <https://clementromac.github.io/>

EDUCATION

University of Bordeaux, Bordeaux *2022 - 2026*
Ph.D. in Computer Science
Pierre-Yves Oudeyer (Inria, FLOWERS) & Thomas Wolf (Hugging Face)

University of Bordeaux, Bordeaux *2019 - 2020*
MSc. in Computer Science
Algorithms and Models specialisation

Ynov Informatique Ingésup, Bordeaux *2014 - 2019*
Degree in Computer Science engineering
Data Science specialisation Rank : 1

High School Les Iris, Lormont *2011 - 2014*
High School Degree, with the highest honours Overall Notation : 17.75/20

WORK EXPERIENCE

Inria (FLOWERS team), Bordeaux *April 2026 - September 2026*
Research Engineer

- Studying curiosity models, LLMs, and automatic curriculum learning for educational technologies.

Hugging Face *January 2022 - January 2026*
Research Scientist

- Studying how to turn LLMs into autonomous RL agents.

Inria (FLOWERS team), Bordeaux *January 2022 - January 2026*
PhD student

- Studying how to turn LLMs into autonomous RL agents.

Inria (FLOWERS team), Bordeaux *September 2020 - December 2021*
Research Engineer

- Working on an experimental platform to apply curiosity-driven algorithms to automatic discovery of Complex Systems (e.g., self-organizing)

Inria (FLOWERS team), Bordeaux *March 2020 - August 2020*
Research intern

- Worked on TeachMyAgent, a benchmark platform for Automatic Curriculum Learning applied to Deep Reinforcement Learning algorithms :
 - Created procedurally generated Box2D environments and embodiments
 - Implemented state-of-the-art Automatic Curriculum Learning methods
 - Built a benchmark setup and benchmarked the methods
 - *Paper accepted at ICML 21'*

Weenove, Bordeaux *October 2017 - February 2020*
Part time Data Scientist

- In charge of the Machine Learning R&D :
 - Implemented an Automated Machine Learning service
 - Made this service available (Beta) in a Business Intelligence Software (Biwee)
 - Lead the Machine Learning R&D projects
 - Data Science projects

Scalian, Bordeaux

Summer 2017

Intern Data Scientist

- Data Science projects :
 - Time Series Anomaly Detection
 - Barcode localization on drone pictures with Deep Learning
 - Data Exploration project

Weenove, Bordeaux

June 2016 - May 2017

Part time Intern Data Scientist

- Machine Learning R&D :
 - Implemented an Automated Machine Learning service
 - Made this service available (Beta) in a Business Intelligence Software (Biwee)

PROJECTS

ChatGPT expliqué en 5 minutes (2023)

A series of short videos explaining Large Language Models. The project can be found **here**.

Lamorel (2022)

Lamorel is a Python library designed for RL practitioners eager to use LLMs. It relies on a distributed client-server architecture, which allows seamless scaling (**repository**).

Automated Discovery Tool (2021)

A software for assisted and automated discovery of patterns in the exploration of complex systems (**repository**).

Interactive Deep RL Demo (2021)

I supervised the project of turning the environments I developed for TeachMyAgent into an on-browser interactive demo of Deep Reinforcement Learning agents' adaptation to unknown 2D tasks. The project can be found **here**.

PUBLICATIONS

MAGELLAN : Metacognitive predictions of learning progress guide autotelic LLM agents in large goal spaces

Loris Gaven, Thomas Carta, Clément Romac, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Forty-Second International Conference on Machine Learning (ICML) 2025*.

WorldLLM : Improving LLMs' world modeling using curiosity-driven theory-making

Guillaume Levy, Cedric Colas, Pierre-Yves Oudeyer, Thomas Carta, Clement Romac. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM) 2025*.

Reinforcement Learning for Aligning Large Language Models Agents with Interactive Environments : Quantifying and Mitigating Prompt Overfitting

Mohamed Salim Aissi, Clement Romac, Thomas Carta, Sylvain Lamprier, Pierre-Yves Oudeyer, Olivier Sigaud, Laure Soulier, Nicolas Thome. *Findings of the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2025*.

SAC-GLAM : Improving Online RL for LLM agents with Soft Actor-Critic and Hindsight Relabeling

Loris Gaven, Clément Romac, Thomas Carta, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Intrinsically Motivated Open-ended Learning Workshop, NeurIPS 2024*.

Grounding Large Language Models in Interactive Environments with Online Reinforcement Learning

Clément Romac*, Thomas Carta*, Thomas Wolf, Sylvain Lamprier, Olivier Sigaud, Pierre-Yves Oudeyer. *Fortieth International Conference on Machine Learning (ICML) 2023*. * equal contribution.

TeachMyAgent : a Benchmark for Automatic Curriculum Learning in Deep RL

Clément Romac* and Rémy Portelas* and Katja Hofmann and Pierre-Yves Oudeyer. *Thirty-eighth International Conference on Machine Learning (ICML) 2021*. * equal contribution.

Meta Automatic Curriculum Learning

Rémy Portelas and Clément Romac and Katja Hofmann and Pierre-Yves Oudeyer. 2020 (Preprint). abs/2011.08463.

Partially Observable Markov Decision Process with Minecraft

Romac et Béraud. 2019 (Preprint). abs/1903.04311.

LANGUAGES

French	Native speaker
English	Advanced (TOEIC : 980/990, 2019)

TEACHING

Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 14h (Lectures and seminars)</i>	<i>October 2025</i>
Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 14h (Lectures and seminars)</i> <i>Reinforcement Learning - 7h (Lectures and seminars)</i>	<i>October 2024</i>
Ecole Nationale Supérieure de Cognitique Bordeaux <i>Last year engineering students from AI Specialisation</i> <i>Transformers and LLMs - 7h (Lectures and seminars)</i>	<i>October 2023</i>
Ynov Informatique Ingésup Bordeaux <i>Masters student, Data Science Specialisation</i> <i>Machine Learning Major - 90h (Lectures and seminars)</i>	<i>November 2018 - December 2018</i>

STUDENTS SUPERVISED

Ryan Boldi <i>Undergrad internship</i>	<i>May-September 2025</i>
Loris Gaven <i>Master's internship</i>	<i>May-September 2024</i>
Guillaume Levy <i>Master's internship</i>	<i>May-September 2024</i>
Matthieu Perie <i>Part-time master's student</i>	<i>September 2021 - September 2023</i>
Paul Germon <i>Master's internship</i>	<i>May-September 2021</i>

SERVICE AND LEADERSHIP

Co-founded the AI working group of Ynov Informatique Ingésup Bordeaux

— We gather every two weeks on Wednesdays to work on AI projects.

Co-founded and animated (until 2019) the meetup “Les nuits des réseaux de neurones”

— Talks and workshops on Deep Learning starting in the evening and continuing throughout the night.

— Held every month or two months, open to everyone (270 members).

— <https://www.meetup.com/fr-FR/Les-nuits-des-reseaux-de-neurones/>

Co-organizer of the meetup “Bordeaux Machine Learning Meetup”

— Talk and workshops once a month (+1400 members)

— <https://www.meetup.com/fr-FR/Bordeaux-Machine-Learning-Meetup/>

I have also been involved in various dissemination initiatives, in particular towards education (I am co-leading the development of the “ChatGPT explained in 5 minutes” series of videos that is used in French classrooms and now embedded in official trainings for workers from all French ministries)

HOBBIES

Sport : In my spare time, I enjoy doing sport, from outdoor activities (hiking, mountaineering, running, trail, canoeing...) to football, tennis padel, badminton...

Music : I have been playing bass guitar for several years, including in a jazz fusion band.